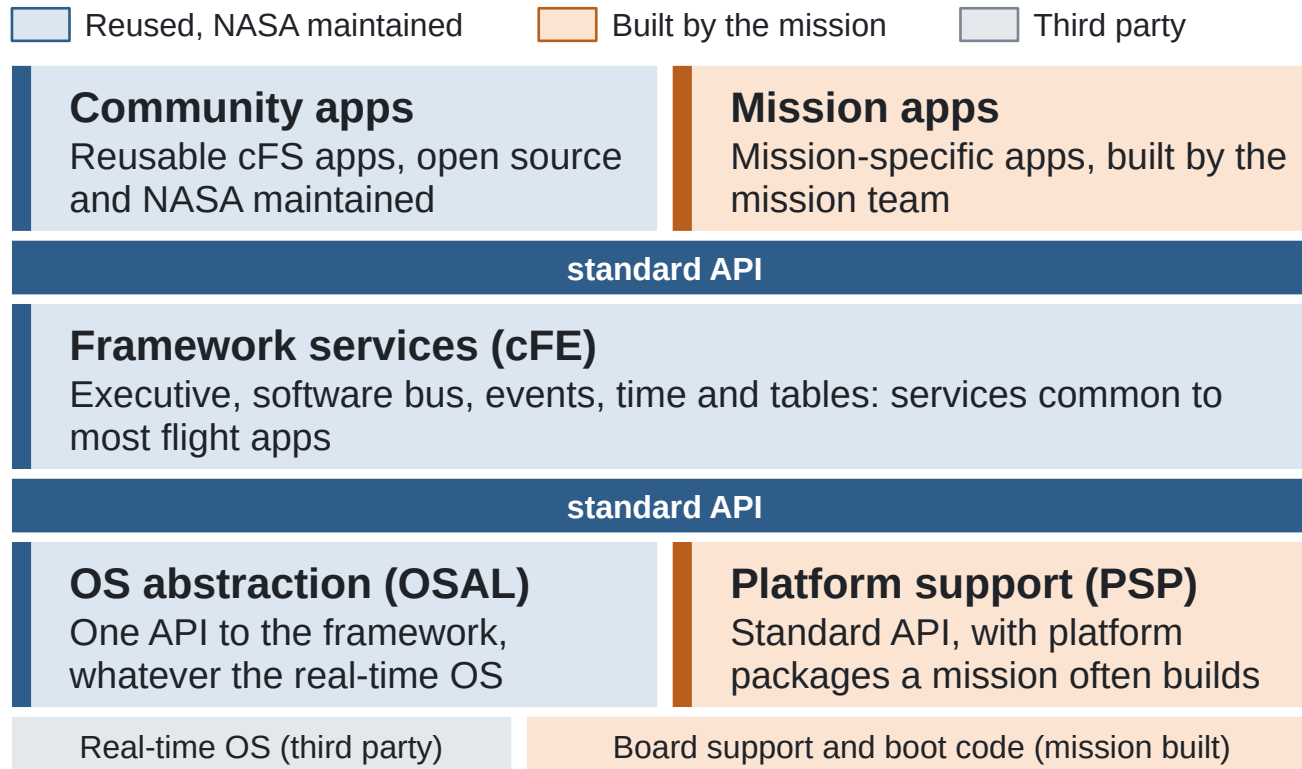


Each layer hides its implementation behind a standard API, so a mission reuses the framework and focuses on custom parts



What NASA says the layers give a mission

Reusable components

“Reuse of tested, certified components supplies savings in each phase of the software development cycle.”

Feature: 2015 deck, p8. Quote: 2021 training memo, p41.

Platform independent

“Layered architecture and compile-time configuration parameters make it scalable and portable to a wide range of platforms”

Feature: 2015 deck, p8. Quote: 2015 deck, p2.

New technology without framework changes

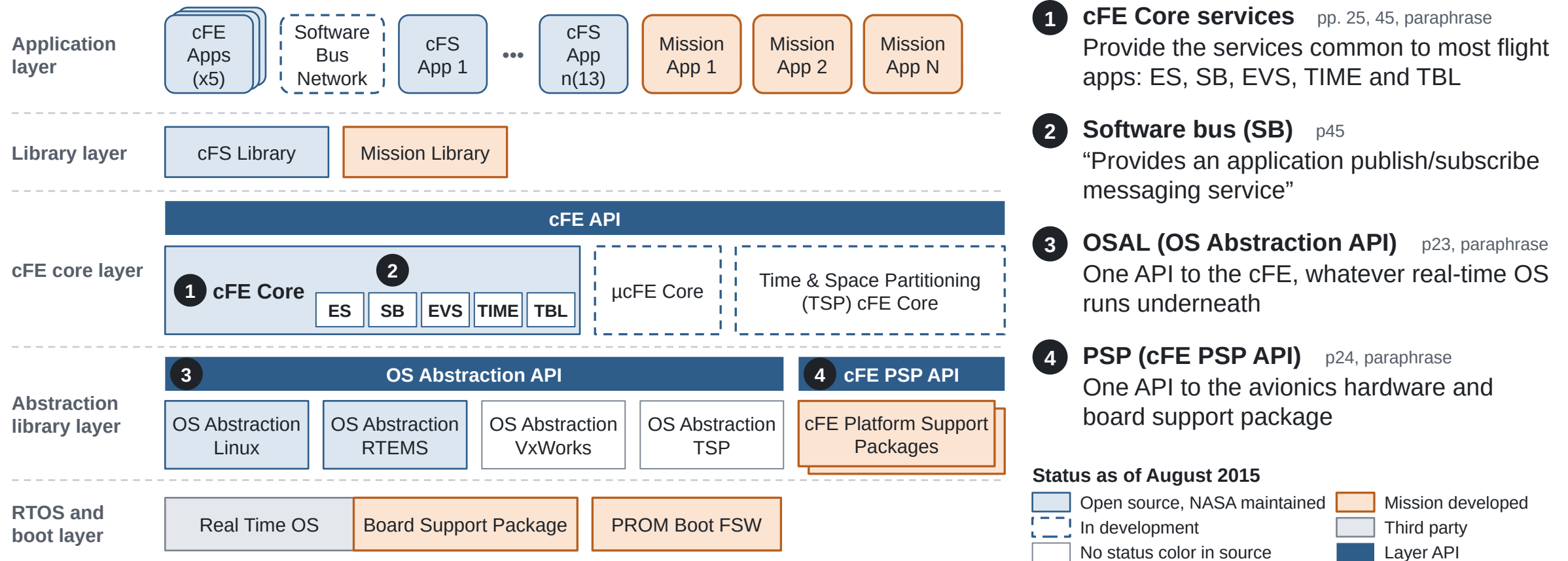
“Internals of a layer can be changed -- without affecting other layers' internals and components.”

Feature: 2015 deck, p8. Quote: 2023 overview, slide 8.

Sources: 2015 deck: McComas et al., NASA Goddard, Aug. 2015, NTRS 20150018075. 2023 overview: Baker, NASA Goddard, 2023, NTRS 20230002444. 2021 memo: NASA/TM-20205000691/REV 1, NTRS 20205011588, printed pages. Title: 2023 overview slide 8 and memo p41. Layers: memo pp. 23 to 26 and 45. Colors: 2015 legend (p8) and its note on center PSPs (p23).

Portfolio reconstruction of a public NASA slide. Not a NASA document.

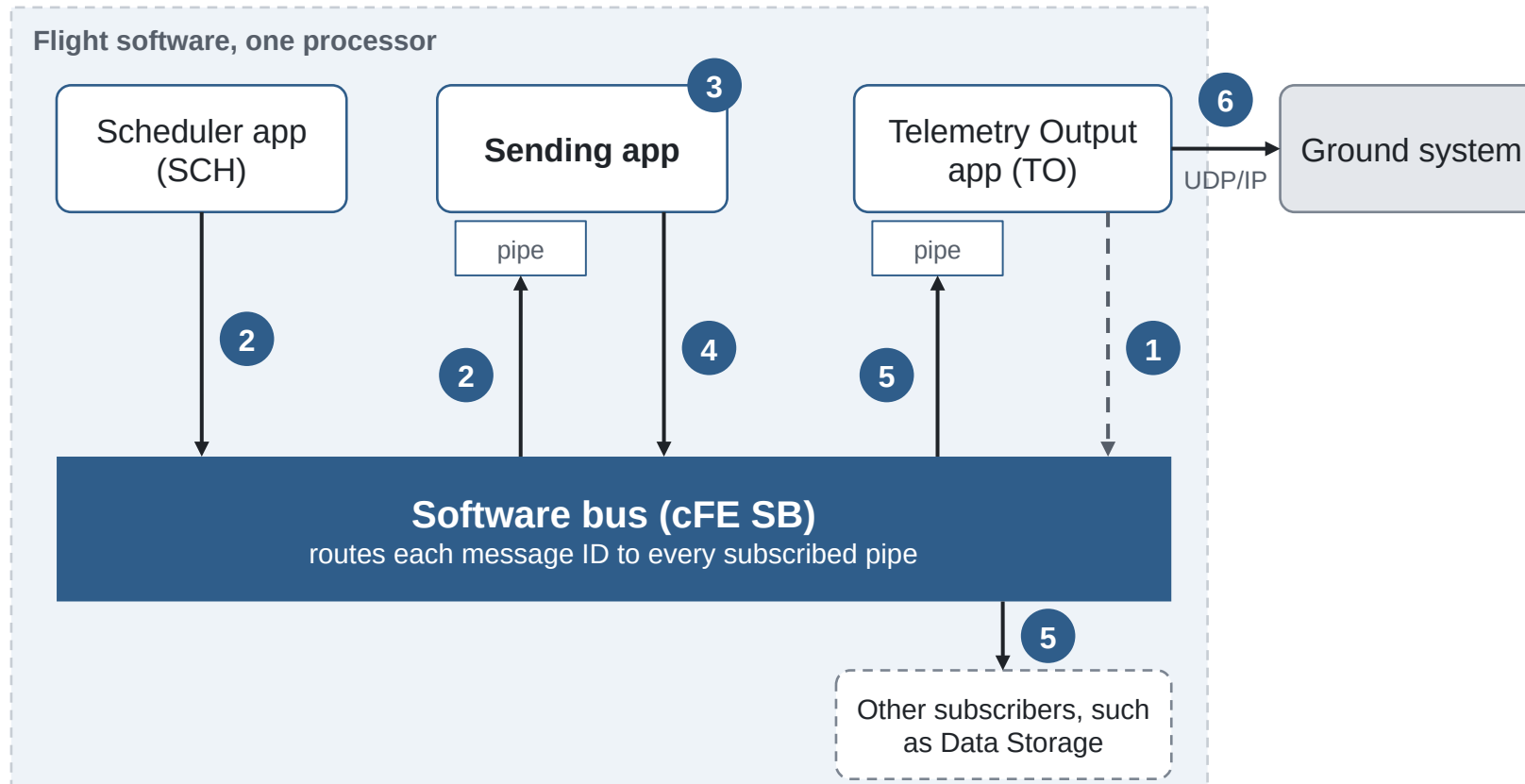
The OSAL and PSP hide the real-time OS and the board, so the cFE and the apps above it stay platform independent



Source: diagram rebuilt from McComas et al., NASA Goddard, Aug. 2015, NTRS 20150018075, p8. Service acronyms and order: 2021 memo, printed p45 (the 2015 deck lists the same five services on p9). Roles: NASA/TM-20205000691/REV 1, Jan. 2021, NTRS 20205011588, printed page on each heading. Dark bars: each layer's standard API (2023 overview, NTRS 20230002444, slide 8).

Portfolio reconstruction of a public NASA slide. Not a NASA document.

One housekeeping packet crosses the software bus in six steps, and the app that sends it does not know who subscribed



- 1 Subscribe.** At startup, TO subscribes a pipe to the packet's ID.
2021 memo, p85 and p93
- 2 Request.** The scheduler sends a housekeeping request over the bus.
2021 memo, p86 and p34
- 3 Build.** The app fills its housekeeping packet and stamps the time.
2021 memo, p88 and p98
- 4 Send.** The app sends the packet without knowing who subscribes.
2021 memo, p85 and p97
- 5 Route.** The bus routes it to every pipe subscribed to its ID.
2021 memo, p85
- 6 Downlink.** TO reads its pipe and sends the packet over UDP/IP.
2021 memo, p95 and p34

Sources: steps from NASA/TM-20205000691/REV 1 (the 2021 memo), Jan. 2021, NTRS 20205011588, printed pages. TO in the open-source bundle is a lab app meant as an example (nasa/cFS README). The memo's CFE_SB_SendMsg and CFE_SB_RcvMsg (p97) are CFE_SB_TransmitMsg and CFE_SB_ReceiveBuffer in the current cFE Application Developers Guide (nasa/cFE, sec. 6.6 and 6.7, read Oct. 6, 2026).

Portfolio diagram drawn from public NASA documents. Not a NASA document.